

Direct Sequence Localization for Distributed Potential Problems

Abstract

We study distributed local search in the deterministic LOCAL model. Balliu, Boudier, d’Amore, Kuhn, Olivetti, Schmid, and Suomela introduced locally optimal problems and showed that every bounded-degree problem governed by a local potential admits a polylogarithmic-round algorithm. Their analysis localizes each improving move separately and then localizes a sequence of moves in an auxiliary interaction graph.

The main observation here is that the whole sequence can instead be replayed inside one ball of the original graph. Minimality pays for the discarded part of every move, and the only remaining error is charged to potential terms crossing the ball boundary. For a radius- ρ potential, normalized local scale K , and normalized upper bound P on the global potential, this gives a deterministic algorithm in

$$O_\rho\left(\left[1 + \min\left\{K(\Delta^\rho + 1) \log n \log(2P), \sqrt{Kn \log n \log(2P)}\right\}\right] \log^2 n \text{ poly}(\log \log n)\right)$$

rounds. Thus, for fixed local scale, the polynomial part is $\min\{\Delta^\rho, \sqrt{n}\}$.

For edge potentials the boundary is sharper. In particular, ordinary locally optimal cut has a deterministic

$$O(\min\{\Delta \log^4 n, \sqrt{n} \log^3 n\} \text{ poly}(\log \log n))$$

algorithm. Together with the randomized and quantum lower bound of Balliu et al., this determines the polynomial dependence on Δ and n up to logarithmic factors. As supporting evidence that the boundary-volume dependence of such generic local-search algorithms is inherent, we also analyze the stronger task in which no subset of a radius- s ball may improve a cut. For every fixed $s \geq 1$, its deterministic upper bound and randomized/quantum lower bound have the same polynomial part

$$\min\{\Delta^{s+1}, \sqrt{\Delta n}, n^{2/3}\}.$$

1 Introduction

In an undirected graph $G = (V, E)$, a cut is a labeling $x : V \rightarrow \{0, 1\}$. It is *locally optimal* if flipping any one vertex does not increase the number of cut edges. Equivalently, every vertex has at least as many neighbors on the other side of the cut as on its own side. A centralized improvement process is immediate: repeatedly flip an improving vertex. Since each flip increases the cut by at least one edge, the process stops after at most $|E|$ flips. The challenge is to parallelize this process without creating a long cascade of mutually dependent repairs.

We work in the standard synchronous LOCAL model [5, 6]: vertices have distinct identifiers, messages have unbounded size, and after T rounds a vertex can know its entire radius- T neighborhood. Local computation is free. The problem of finding a locally optimal cut will be abbreviated by LOC.

Balliu et al. [2] introduced a broad framework of locally optimal problems (LOPs). A global potential is written as a sum of bounded-radius local terms, and an invalid vertex has a one-block relabeling that strictly decreases this potential. They obtained the first polylogarithmic-round algorithms for all bounded-degree LOPs. For LOC, their deterministic complexity is $O(\Delta^2 \log^8 n \text{ poly}(\log \log n))$, and they proved an $\Omega(\min\{\Delta, \sqrt{n}\})$ lower bound for randomized and quantum-LOCAL algorithms in the usual range $\Delta \leq cn$, for any fixed $c < 1$.

The present work improves the upper bound by changing the localization argument. The prior proof first replaces a large improving move by a small-diameter move and then represents moves as vertices of an auxiliary interaction graph. The two localization steps multiply their losses. Here the entire sequence is replayed directly inside a ball of G . The proof counts occurrences of original vertices, so the auxiliary graph is unnecessary.

1.1 What is new and what is inherited

The definitions of LOP, improving sets, improving ratios, and minimal improving sets, as well as the threshold-scheduling paradigm, originate in Balliu et al. [2]. We also use the deterministic strong-diameter network decomposition of Ghaffari and Grunau [4]; this line of work began with the polylogarithmic deterministic decomposition of Rozhoň and Ghaffari [7].

The new ingredients and conclusions are:

1. a direct replay lemma for a whole sequence of minimal improving moves;
2. a one-step sequence-localization theorem in the original graph;
3. a deterministic generic LOP algorithm whose polynomial dependence improves from $\Delta^{2\rho}$ to $\min\{\Delta^\rho, \sqrt{n}\}$ at fixed local scale;
4. an edge-boundary specialization giving a deterministic $\tilde{O}(\min\{\Delta, \sqrt{n}\})$ algorithm for LOC, which matches the known lower bound up to logarithmic factors; and
5. a supporting upper/lower-bound pair for radius- s local cut. The latter is not needed for the LOC algorithm; its role is to show that the polynomial boundary-volume scales encountered by generic local-search methods occur on concrete simple unweighted graphs.

The distinction is summarized below. The table suppresses logarithmic factors and fixed-radius constants.

Problem	Previously known	Result here
General radius- ρ LOP	$\tilde{O}(K^2 \Delta^{2\rho})$ deterministic	$\tilde{O}_\rho(\min\{K \Delta^\rho, \sqrt{Kn}\})$ deterministic
Ordinary LOC	$\tilde{O}(\Delta^2)$ deterministic	$\tilde{O}(\min\{\Delta, \sqrt{n}\})$ deterministic
Radius- s local cut	–	$\tilde{\Theta}_s(\min\{\Delta^{s+1}, \sqrt{\Delta n}, n^{2/3}\})$

The $\tilde{\Theta}_s$ notation in the last row compares a deterministic upper bound with a randomized and quantum lower bound, and hence refers only to the polynomial dependence on n and Δ .

2 Model and normalized potential parameters

For $A \subseteq V$, let $N_t[A] = \{u : \text{dist}(u, A) \leq t\}$, and write $N_t[v] = N_t[\{v\}]$. A radius- ρ local-potential problem has a finite output alphabet and a potential

$$\text{Pot}_G(\ell) = \sum_{z \in V} \phi_z(\ell|_{N_\rho[z]}), \quad 0 \leq \phi_z \leq \Lambda. \quad (1)$$

The input labels and the topology inside $N_\rho[z]$ may be arguments of ϕ_z ; we suppress them. A “vertex block” may contain the output of a vertex and its incident half-edge outputs. Thus this notation includes the formal LOP model of [2]; all that is used below is that legal block relabelings are finite and restrictions of a legal multi-block relabeling are legal.

We assume that whenever the current output is invalid at a vertex, some relabeling of its block decreases (1) by at least $\lambda > 0$. Divide all potentials by λ and define

$$K = \frac{\Lambda}{\lambda}, \quad P \geq \max_\ell \frac{\text{Pot}_G(\ell)}{\lambda}. \quad (2)$$

Henceforth potentials are normalized so that the guaranteed correction is at least one, each local term lies in $[0, K]$, and every global potential is at most P . The algorithm only needs a common upper bound on P . The universal choice $P = Kn$ is available for (1); retaining P makes the dependence on the maximum global potential explicit. The factor K is the one unavoidable local scale parameter. For a fixed problem family at bounded degree, it is a constant.

Definition 2.1 (Improving move). For a labeling ℓ , a move $M = (A, \ell_A)$ consists of a nonempty support $A \subseteq V$ and a legal relabeling supported on A . If ℓ^M is the resulting labeling, define

$$\text{Imp}_\ell(M) = \text{Pot}_G(\ell) - \text{Pot}_G(\ell^M), \quad \text{IR}_\ell(M) = \frac{\text{Imp}_\ell(M)}{|A|}.$$

The move is improving if $\text{Imp}_\ell(M) > 0$. An improving move of ratio q is *minimal* if every move supported on a proper nonempty subset of A has ratio at most q .

Every improving move contains a minimal improving move of at least the same ratio: choose, among all improving moves supported inside its support, one of maximum ratio and then inclusion-minimal support. A sequence $M_1, \dots, M_h$ is β -improving if M_i is evaluated after the preceding moves and has ratio at least β .

Let

$$B_\rho(\Delta) = 1 + \Delta \sum_{j=0}^{\rho-1} (\Delta - 1)^j = O_\rho(\Delta^\rho) \quad (3)$$

be the maximum size of a radius- ρ ball in a graph of maximum degree Δ .

3 Direct localization of an improving sequence

Fix an initial labeling ℓ_0 and a sequence of minimal β -improving moves $M_i = (A_i, \ell_i)$, where ℓ_i is the labeling after move i . Count repeated changes by

$$w(u) = |\{i : u \in A_i\}|, \quad w(U) = \sum_{u \in U} w(u).$$

The sequence decreases the normalized potential by at least $\beta w(V)$, while the initial potential is at most P . Therefore

$$w(V) \leq \frac{P}{\beta}. \quad (4)$$

For $U \subseteq V$, the *replay inside U* starts from ℓ_0 , follows the labels of the true sequence on U , and permanently freezes all blocks outside U at their labels in ℓ_0 . Thus replay step i applies only $A_i \cap U$.

Lemma 3.1 (Direct replay). *Let ℓ'_h be the final labeling obtained by replaying a minimal β -improving sequence inside U . Then*

$$\text{Pot}_G(\ell_0) - \text{Pot}_G(\ell'_h) \geq \beta w(U) - 2KB_\rho(\Delta) \sum_{\substack{u \in U \\ \text{dist}(u, V \setminus U) \leq 2\rho}} w(u).$$

More precisely, every occurrence of a boundary vertex can be charged only for local terms centered in $N_\rho[u]$, and each such term contributes discrepancy at most $2K$.

Proof. Consider step i and put $B = A_i \cap U$ and $C = A_i \setminus U$. Let q_i be the ratio of the full move. In the true state immediately before step i , first apply the prescribed part on C . Minimality says that its improvement is at most $q_i|C|$; this is also true if the improvement is nonpositive. Since the full move improves by $q_i(|B| + |C|)$, the subsequent prescribed relabeling on B has marginal improvement at least $q_i|B|$.

Compare this marginal with replay step i . Before and after the step, the two worlds agree on U . A local term can have different marginals only if its radius- ρ scope contains both a changed vertex of B and a vertex outside U . Such a changed vertex lies within distance 2ρ of $V \setminus U$. A fixed occurrence of u belongs to at most $B_\rho(\Delta)$ local scopes, and the difference between two marginals of one term in $[0, K]$ is at most $2K$. Hence replay step i improves by at least

$$q_i|B| - 2KB_\rho(\Delta)|\{u \in B : \text{dist}(u, V \setminus U) \leq 2\rho\}|.$$

Sum over i and telescope the replay potential. Since $q_i \geq \beta$, the first sum is at least $\beta w(U)$, and the second sum is exactly the asserted occurrence count. $\square$

The proof intentionally overcounts a local term when several changed vertices affect it. Counting possible centers as well as changed occurrences gives a square-root alternative.

Theorem 3.2 (Sequence localization). *Let $0 < \varepsilon < \beta$, and suppose a vertex v occurs in a sequence of minimal β -improving moves. In the initial labeling there is a minimal improving move of ratio at least $\beta - \varepsilon$, contained in the union of the sequence and in a ball around v of radius*

$$L = O_\rho \left(1 + \min \left\{ \frac{K(\Delta^\rho + 1)}{\varepsilon} \log \left(1 + \frac{P}{\beta} \right), \sqrt{\frac{Kn}{\varepsilon} \log \left(1 + \frac{P}{\beta} \right)} \right\} \right). \quad (5)$$

No diameter assumption on the moves in the sequence is required.

Proof. Let $U_i = N_{2\rho i}[v]$, $W_i = w(U_i)$, and for $i \geq 1$ put

$$a_i = W_i - W_{i-1}, \quad Z_i = |U_{i+1} \setminus U_{i-1}|.$$

Because v occurs, $W_0 \geq 1$. For every k ,

$$\sum_{i=1}^k \frac{a_i}{W_i} \leq \sum_{i=1}^k \log \frac{W_i}{W_{i-1}} = \log \frac{W_k}{W_0} \leq \log \left(1 + \frac{P}{\beta} \right), \quad (6)$$

where the last inequality uses (4). The annuli $U_{i+1} \setminus U_{i-1}$ overlap at most twice, so

$$\sum_{i=1}^k Z_i \leq 2n. \quad (7)$$

Replay inside U_i . Any changed vertex within distance 2ρ of the complement belongs to $U_i \setminus U_{i-1}$, whose total occurrence mass is a_i . The center of every discrepant local term lies in $U_{i+1} \setminus U_{i-1}$. Repeating the proof of Lemma 3.1, but using the smaller of the changed-vertex and center counts, shows that the replay loses at most

$$2Ka_i \min\{B_\rho(\Delta), Z_i\}. \quad (8)$$

Suppose the quantity in (8) exceeds εW_i for every $1 \leq i \leq k$. Using the first term in the minimum and (6) gives

$$k = O_\rho \left(\frac{K(\Delta^\rho + 1)}{\varepsilon} \log \left(1 + \frac{P}{\beta} \right) \right).$$

Using the second term, then Cauchy–Schwarz and (7), gives

$$\log\left(1 + \frac{P}{\beta}\right) \geq \frac{\varepsilon}{2K} \sum_{i=1}^k \frac{1}{Z_i} \geq \frac{\varepsilon k^2}{4Kn}.$$

If some $Z_i = 0$, the boundary error is already zero. Consequently, within the number of annuli displayed in (5), some i has boundary loss at most εW_i .

For this i , replay improves the initial potential by at least $(\beta - \varepsilon)W_i > 0$. Let A be the set of blocks on which the final replay differs from ℓ_0 . Every vertex of A occurs at least once, hence $|A| \leq W_i$; the replay move has ratio at least $\beta - \varepsilon$. Taking a minimal improving submove cannot decrease the ratio. Its support is contained in U_i and in the union of the original sequence. $\square$

4 The generic deterministic algorithm

We use a strong-diameter network decomposition: a coloring of clusters such that adjacent clusters have different colors and every cluster has bounded diameter in its induced subgraph. Ghaffari and Grunau [4] give, in $\tilde{O}(\log^2 n)$ deterministic rounds, a decomposition with $Q = O(\log n)$ colors and strong diameter $D = O(\log n)$.

Set

$$\eta = \frac{1}{8Q}, \quad R_j = \frac{1}{4} + \frac{j-1}{4Q}, \quad 1 \leq j \leq Q+1.$$

Then $R_{j+1} - \eta > R_j$ and $R_{Q+1} = 1/2$. Choose L from Theorem 3.2 uniformly for $\beta \geq 1/4$ and $\varepsilon = \eta$, and let $p = 2(L + \rho) + 1$.

Compute a network decomposition of the power graph G^p . For a cluster C , let its write region be $N_L[C]$ and its observation region be $N_{L+2\rho}[C]$. Distinct clusters of one color have write regions at distance greater than 2ρ , so no local potential term is affected by two of them. A cluster has diameter at most pD when power-graph edges are replaced by paths in G ; its observation region can therefore be gathered at a leader in $O(pD + L + \rho)$ rounds.

Process the Q colors in order. In a color- j cluster, the leader repeatedly chooses a maximum-ratio improving relabeling supported in the write region and applies it while its ratio is at least R_j . The leader can brute-force this finite optimization because local computation is unbounded. A maximum-ratio choice is minimal. The simulation terminates because the label space is finite and every accepted move strictly decreases the potential. Same-color simulations are independent and may be serialized for analysis.

Theorem 4.1 (Generic deterministic upper bound). *Every normalized radius- ρ local-potential problem described in Section 2 can be solved deterministically in*

$$O_\rho\left(\left[1 + \min\left\{K(\Delta^\rho + 1) \log n \log(2P), \sqrt{Kn \log n \log(2P)}\right\}\right] \log^2 n \text{ poly}(\log \log n)\right) \quad (9)$$

LOCAL rounds. If $K = O(1)$ and $P \leq n^{O(1)}$, this is $\tilde{O}_\rho(\min\{\Delta^\rho, \sqrt{n}\})$.

Proof. It remains to prove correctness and count rounds. Suppose the final labeling is invalid at v , and let C be the decomposition cluster containing v , of color j . Consider the state just after color j was processed. Serialize every move made in later color phases and append, in the final state, a correcting singleton move at v . Every actual later move is minimal and has ratio at least R_{j+1} ; the appended correction has ratio at least one and is minimal. Thus these moves form a minimal R_{j+1} -improving sequence in which v occurs.

Apply Theorem 3.2 to this sequence. In the state immediately after color j , it produces a move of ratio at least $R_{j+1} - \eta > R_j$ supported in $N_L[v] \subseteq N_L[C]$. This contradicts exhaustion of the write region during phase j . Hence the final labeling is valid.

One round on G^p costs at most p rounds on G . The decomposition costs $p\tilde{O}(\log^2 n)$ rounds. The Q color phases cost $O(Q(pD + L + \rho)) = O(p \log^2 n)$ rounds. Substituting $\varepsilon = \Theta(1/\log n)$ and $\beta \geq 1/4$ in (5) yields (9). $\square$

Remark 4.2 (Comparison with prior work). The prior deterministic bound of Balliu et al. [2] has polynomial part $K^2 \Delta^{2\rho}$. Theorem 4.1 removes one localization step, one power of the local scale, and one power of the radius- ρ ball volume. It also exposes the alternative $\sqrt{Kn}$ scale. No matching lower bound is asserted for every abstract LOP; the tight statement below is for the concrete edge-potential problem LOC.

5 Edge potentials and a tight algorithm for locally optimal cut

Suppose the potential is a sum over edges,

$$\text{Pot}_G(\ell) = \sum_{e \in E} \psi_e(\ell|_e), \quad 0 \leq \psi_e \leq \Gamma.$$

After division by the minimum correction λ , put $K_E = \Gamma/\lambda$ and again let P bound the normalized global potential. Such problems are called locally optimal problems on edges (LOPEs) in [2].

Lemma 5.1 (Edge replay). *For a sequence replayed inside U ,*

$$\text{Pot}_G(\ell_0) - \text{Pot}_G(\ell'_h) \geq \beta w(U) - 2K_E \sum_{u \in U} w(u) \deg_{V \setminus U}(u).$$

Proof. Repeat the proof of Lemma 3.1. A marginal can differ between the true and frozen-outside executions only on an edge with one changed endpoint in U and the other endpoint outside U . The discrepancy per crossing edge is at most $2K_E$. Charging it to the changed endpoint proves the formula. $\square$

Take $U_i = N_i[v]$, let a_i be the occurrence mass of the i th layer, and put $Z_i = |U_{i+1} \setminus U_{i-1}|$. A changed boundary vertex has at most $\min\{\Delta, Z_i\}$ neighbors outside U_i . The proof of Theorem 3.2 therefore gives an edge-potential localization radius

$$O\left(1 + \min\left\{\frac{K_E \Delta}{\varepsilon} \log\left(1 + \frac{P}{\beta}\right), \sqrt{\frac{K_E n}{\varepsilon} \log\left(1 + \frac{P}{\beta}\right)}\right\}\right). \quad (10)$$

Using this radius in the scheduler from Section 4 proves the following edge version of Theorem 4.1.

Theorem 5.2 (Edge-potential upper bound). *A normalized edge-potential problem can be solved deterministically in*

$$O\left(\left[1 + \min\left\{K_E \Delta \log n \log(2P), \sqrt{K_E n \log n \log(2P)}\right\}\right] \log^2 n \text{ poly}(\log \log n)\right)$$

rounds.

For LOC, use the potential

$$\text{Pot}_G(x) = |\{uv \in E : x(u) \neq x(v)\}|,$$

the number of uncut edges. If v is invalid, flipping v decreases the potential by a positive integer, hence by at least one. Thus $K_E = 1$, and the maximum global potential is $P = |E| \leq n^2/2$.

Corollary 5.3 (Tight deterministic algorithm for LOC). *Locally optimal cut can be solved deterministically in*

$$O\left(\min\{\Delta \log^4 n, \sqrt{n} \log^3 n\} \text{poly}(\log \log n)\right)$$

rounds.

Balliu et al. [2, Theorem 5.2 and Corollary 5.6] proved that, for every fixed $c < 1$ and $\Delta \leq cn$, randomized and quantum-LOCAL algorithms with constant global success probability require $\Omega(\min\{\Delta, \sqrt{n}\})$ rounds. Corollary 5.3 therefore settles the polynomial dependence on both parameters. This tightness claim does not include the remaining logarithmic factors.

6 A supporting barrier: radius-local cuts

This section is logically independent of the generic algorithm. It studies a stronger cut-stability notion in order to test whether the boundary-volume scales behind direct localization can be improved for all local-search tasks.

Definition 6.1 (Radius- s locally optimal cut). For a fixed integer $s \geq 1$, a cut $x : V \rightarrow \{0, 1\}$ is *radius- s locally optimal* if, for every $v \in V$ and every $S \subseteq N_s[v]$, simultaneously flipping all vertices of S does not increase the number of cut edges.

Ordinary LOC corresponds to allowing only singleton flips, while radius- s optimality quantifies over every coalition contained in one radius- s ball.

Theorem 6.2 (Radius-local cut bounds). *For every fixed $s \geq 1$, radius- s locally optimal cut has a deterministic algorithm with complexity*

$$O_s\left(\min\{\Delta^{s+1} \log^4 n, \sqrt{\Delta n} \log^3 n, n^{2/3} \log^{8/3} n\} \text{poly}(\log \log n)\right). \quad (11)$$

Conversely, for all sufficiently large feasible parameter values, every randomized or quantum-LOCAL algorithm with global success probability at least $2/3$ requires

$$\Omega_s\left(\min\{\Delta^{s+1}, \sqrt{\Delta n}, n^{2/3}\}\right) \quad (12)$$

rounds, even on connected simple unweighted graphs.

The upper bound is included for completeness, but the main message for generic algorithms is the lower bound. In the degree-dominated regime $\Delta \leq n^{1/(2s+1)}$, the $\Omega_s(\Delta^{s+1})$ term exactly matches the volume scale of an edge interaction combined with a radius- s correction. Thus no algorithm that solves all such local-search instances can improve this degree exponent. This is evidence about the limits of generic methods, not a claim that every abstract LOP individually satisfies this lower bound.

6.1 Weighted normalization and the upper bound

For each vertex define

$$b(u) = \max_{z \in N_s[u]} |N_s[z]|, \quad c(u) = \frac{1}{b(u)}, \quad c(S) = \sum_{u \in S} c(u). \quad (13)$$

These values are computable from radius- $2s$ neighborhoods.

Lemma 6.3 (A permitted correction has unit cost). *If $S \subseteq N_s[v]$, then $c(S) \leq 1$.*

Proof. For $u \in N_s[v]$, undirectedness gives $v \in N_s[u]$, and hence $b(u) \geq |N_s[v]|$. Therefore $c(S) \leq |S|/|N_s[v]| \leq 1$. $\square$

For a nonempty flip S , define its weighted improving ratio as its cut improvement divided by $c(S)$. A positive move is weighted minimal if no submove has larger weighted ratio. Consider a sequence of weighted minimal moves of ratio at least β . If $w(u)$ counts occurrences, write

$$\mu(u) = c(u)w(u), \quad W(U) = \sum_{u \in U} \mu(u).$$

Since the cut potential ranges over an interval of length at most $|E|$,

$$\beta W(V) \leq |E|. \quad (14)$$

If v occurs, then $W(\{v\}) \geq c(v) \geq 1/n$.

Lemma 6.4 (Weighted replay). *Replaying the sequence inside U decreases the uncut-edge potential by at least*

$$\beta W(U) - 2 \sum_{u \in U} \mu(u)b(u) \deg_{V \setminus U}(u).$$

Proof. At one step, split the full support into $A \subseteq U$ and $B \subseteq V \setminus U$. If the full weighted ratio is q , weighted minimality bounds the improvement of B alone by $qc(B)$. Hence applying A after B gains at least $qc(A)$. Freezing the exterior changes the marginal only on crossing edges, with discrepancy at most two per edge. Sum over steps and use $w(u) = \mu(u)/c(u) = \mu(u)b(u)$. $\square$

Lemma 6.5 (Weighted sequence localization). *Let $0 < \varepsilon < \beta$ and suppose v occurs in a weighted minimal β -improving sequence. In the initial cut there is a weighted minimal move of ratio at least $\beta - \varepsilon$, supported in the sequence union and within distance*

$$O_s \left(1 + \min \left\{ \frac{\Delta(\Delta^s + 1)}{\varepsilon} \log \left(1 + \frac{n^3}{\beta} \right), \sqrt{\frac{\Delta n}{\varepsilon} \log \left(1 + \frac{n^3}{\beta} \right)}, \left(\frac{n^2}{\varepsilon} \log \left(1 + \frac{n^3}{\beta} \right) \right)^{1/3} \right\} \right) \quad (15)$$

of v .

Proof. Set $h = 2s + 1$, $U_i = N_{hi}[v]$, $W_i = W(U_i)$, $a_i = W_i - W_{i-1}$, and $Z_i = |U_{i+1} \setminus U_{i-1}|$. A boundary vertex $u \in U_i$ lies in $U_i \setminus U_{i-1}$. If it has a neighbor outside U_i , then for every $z \in N_s[u]$, the ball $N_s[z]$ lies inside $U_{i+1} \setminus U_{i-1}$. Consequently

$$b(u) \leq \min\{O_s(\Delta^s), Z_i\}, \quad \deg_{V \setminus U_i}(u) \leq \min\{\Delta, Z_i\}.$$

Lemma 6.4 bounds the replay error by

$$O_s(a_i) \min\{\Delta(\Delta^s + 1), \Delta Z_i, Z_i^2\}. \quad (16)$$

By (14), $W(V) \leq n^2/(2\beta)$, while $W_0 \geq 1/n$. Hence

$$\sum_{i=1}^k \frac{a_i}{W_i} \leq \log \frac{W_k}{W_0} \leq \log \left(1 + \frac{n^3}{\beta} \right). \quad (17)$$

Also $\sum_{i=1}^k Z_i \leq 2n$. If (16) exceeded εW_i for every $i \leq k$, its three terms would give, respectively,

$$\begin{aligned} \log \left(1 + \frac{n^3}{\beta} \right) &= \Omega_s \left(\frac{\varepsilon k}{\Delta(\Delta^s + 1)} \right), \\ \log \left(1 + \frac{n^3}{\beta} \right) &= \Omega \left(\frac{\varepsilon}{\Delta} \sum_{i=1}^k \frac{1}{Z_i} \right) = \Omega \left(\frac{\varepsilon k^2}{\Delta n} \right), \end{aligned}$$

and, using convexity of $z \mapsto z^{-2}$,

$$\log \left(1 + \frac{n^3}{\beta} \right) = \Omega \left(\varepsilon \sum_{i=1}^k \frac{1}{Z_i^2} \right) = \Omega \left(\frac{\varepsilon k^3}{n^2} \right).$$

Thus a good annulus occurs within (15). Replaying there gains at least $(\beta - \varepsilon)W_i$. Its final support A satisfies $c(A) \leq W_i$, so the replay has weighted ratio at least $\beta - \varepsilon$; taking a weighted minimal submove finishes the proof. $\square$

To obtain (11), use the scheduler of Section 4 with $\eta = 1/(8Q)$ and $R_j = 1/4 + (j-1)/(4Q)$. A violated radius- s constraint has an integer cut improvement at least one and, by Lemma 6.3, weighted ratio at least one. The write region for a cluster is enlarged from $N_L[C]$ to $N_{L+s}[C]$. Compute the decomposition on G^p for $p = 2(L+s+1) + 1$ and gather $N_{L+s+2}[C]$ at the leader. Thus distinct same-color write regions are at distance greater than one, and no edge potential is affected by two simultaneous cluster simulations. If a final improving coalition lies in $N_s[v]$, choose a vertex of that coalition and apply Lemma 6.5; the localized move lies in the enlarged write region and contradicts exhaustion of the appropriate color phase. Substituting $\varepsilon = \Theta(1/\log n)$ in (15) and multiplying by the $O(\log^2 n \text{ poly}(\log \log n))$ decomposition overhead gives (11).

6.2 The lower bound

The construction adapts the causal indistinguishability strategy used for LOC in Section 5 and Appendix B of Balliu et al. [2]. We give the graph construction and parameter optimization in full.

Lemma 6.6 (Adjustable expanding core). *For every fixed $s \geq 1$, there is $q_0(s)$ such that, for all integers $q \geq q_0(s)$ and $q \leq M \leq q^s$, there is a simple bipartite graph C with parts of size M , maximum degree at most $2q$, and*

$$|\partial_C X| \geq \frac{q}{4}|X| \quad \text{for every } 0 < |X| \leq M.$$

Here $\partial_C X$ is the set of core edges with exactly one endpoint in X .

Proof. Include each possible bipartite edge independently with probability q/M . Every degree has expectation q , so a Chernoff bound and a union bound give $\Pr[\Delta(C) > 2q] \leq 2Me^{-q/3}$. If X contains a vertices of one side and b of the other, with $t = a + b \leq M$, then

$$\mathbb{E}|\partial_C X| = \frac{q}{M}(a(M-b) + b(M-a)) = q \left(t - \frac{2ab}{M} \right) \geq \frac{qt}{2}.$$

Another Chernoff bound gives $\Pr[|\partial_C X| < qt/4] \leq e^{-qt/16}$. There are at most $\binom{2M}{t} \leq (2eM/t)^t$ choices of size t . Since $\log M \leq s \log q = o(q)$ for fixed s , the sum of all failure probabilities is less than one for sufficiently large q . $\square$

Turn this core into a radius- s gadget. Index the vertices on each side by M distinct words from $[q]^s$ and attach the trie of these words to a new root, with the bipartition side included in the first branch. The root has degree at most $2q$, other internal trie vertices have degree at most $q + 1$, every core vertex is at distance exactly s from the root, and the gadget has $\Theta_s(M)$ vertices. Let $a = \lfloor q/16 \rfloor$. Interfaces will use only side-0 core vertices and will give each such vertex at most $2a + 1$ interface edges. Including its tree-parent edge, a core vertex has fewer than $q/4$ neighbors outside the core when q is large.

Lemma 6.7 (Rigidity). *In every radius- s locally optimal cut, a gadget core is labeled by one of its two proper bipartite colorings.*

Proof. Let $\sigma(u) \in \{0, 1\}$ denote the bipartition side of a core vertex and call $x(u) \oplus \sigma(u)$ its phase. If phases are not constant, let X be the smaller phase class, so $0 < |X| \leq M$. Every edge of $\partial_C X$ is currently uncut and becomes cut when X is flipped. At most $(2a + 2)|X|$ non-core edges can be lost. By Lemma 6.6,

$$|\partial_C X| \geq (q/4)|X| > (2a + 2)|X|$$

for large q . The move is permitted because the entire core lies in the radius- s ball of the root, contradicting local optimality. $\square$

The common label on side-0 core vertices is the gadget phase. Flipping every vertex of a gadget is also a permitted radius- s move. It changes no internal edge and toggles precisely the interface edges.

Number the side-0 vertices modulo M . For every shift $t \in \{0, \dots, a - 1\}$, the edges $j \leftrightarrow j + t \pmod{M}$ form a perfect matching between two gadgets. The a matchings are edge-disjoint. Hence every integer interface weight at most aM is realized by taking a prefix of these edges, with interface degree at most a per port vertex.

Let $k = \lfloor aM/2 \rfloor$. Take gadgets

$$A_0, \dots, A_{k-1}, \quad B_0, \dots, B_{k-1}, \quad Z.$$

For $1 \leq i \leq k - 1$, join A_{i-1} to A_i and B_{i-1} to B_i by interfaces of weight $2i$. Form two inputs:

- in G_{odd} , join A_{k-1} to B_{k-1} by an interface of weight $2k$ and add one edge from A_{k-1} to Z ;
- in G_{even} , join each of A_{k-1} and B_{k-1} to Z by an interface of weight $2k$.

The graphs are connected, simple, and have maximum degree at most $3q$ for large q .

Lemma 6.8 (Forced parity). *Let u and v be side-0 representatives in A_0 and B_0 , respectively. Every radius- s locally optimal cut satisfies*

$$x(u) \neq x(v) \quad \text{in } G_{\text{odd}}, \quad x(u) = x(v) \quad \text{in } G_{\text{even}}.$$

Proof. By Lemma 6.7, every gadget has a phase. On either arm, an inward interface has weight two larger than the preceding outward interface. If the inward interface were uncut, flipping the whole gadget would gain its full weight and lose at most the outward weight, a strict improvement. At the exceptional tip of G_{odd} , the one extra edge can reduce the gain by only one. Hence all main-chain interfaces are cut.

The path of interfaces from A_0 to B_0 has $2k - 1$ links in G_{odd} and $2k$ links in G_{even} . Side-0 labels equal gadget phases, and a cut interface joins opposite phases. The stated parity follows. $\square$

Give corresponding vertices the same identifiers in the two inputs. The graphs differ only at the inner tips and at Z . Any path from u or v to a differing edge crosses $\Omega_s(k)$ gadget interfaces. Therefore, for $T \leq c_s k$, the joint radius- T input neighborhoods of (u, v) are identical in the two graphs. The standard causal indistinguishability argument applies to randomized LOCAL, and also to quantum-LOCAL with prior entanglement independent of the input topology; see [3, 1, 2]. The joint output distribution at (u, v) is the same on both inputs. If it assigns probability p to equality, global success is at most p on G_{even} and at most $1 - p$ on G_{odd} . One input therefore has success probability at most $1/2$. Consequently

$$T = \Omega_s(k) = \Omega_s(qM). \quad (18)$$

There are $\Theta(qM)$ gadgets, each with $\Theta_s(M)$ vertices, so the construction uses $\Theta_s(qM^2)$ vertices. It may be padded to any larger target size by a path attached at Z , identically in the two inputs. Choose a sufficiently small constant multiple

$$q = \Theta_s(\min\{\Delta, n^{1/3}\}), \quad M = \min\left\{q^s, \left\lfloor \sqrt{\frac{n}{C_s q}} \right\rfloor\right\}.$$

For sufficiently large feasible parameters, $q \leq M \leq q^s$, the construction fits in n vertices, and its degree is at most Δ . Equation (18) becomes

$$T = \Omega_s\left(\min\{\Delta^{s+1}, \sqrt{\Delta n}, n^{2/3}\}\right),$$

which proves (12) and completes Theorem 6.2.

7 Discussion

The direct replay argument isolates the source of the speedup: a sequence of local improvements should be localized as a sequence, in the original graph, rather than as a collection of vertices in a second graph. Minimality controls the discarded portion of every move. Once that cost is paid, only an annular boundary remains, and the two ways of counting that boundary produce the degree and square-root branches of the running time.

For ordinary locally optimal cut, edge boundaries make the resulting algorithm polynomially optimal in both Δ and n . For general local potentials, Theorem 4.1 is an upper bound rather than a universal characterization: special algebraic structure may permit faster algorithms. The radius-local cut construction shows, however, that larger correction neighborhoods genuinely realize the boundary-volume exponents and that no generic algorithm covering the entire class can avoid them.

Three questions remain open. First, the logarithmic factors in Corollary 5.3 are not known to be tight. Second, at constant degree there is still a gap between deterministic, randomized, and quantum complexities. Third, it would be useful to identify structural conditions on a potential—beyond being an edge sum—that replace the generic local scale K by a smaller intrinsic boundary parameter.

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